Drafted by Team LifeLink

Multi-Stakeholder Healthcare System Survey & Insights Report

Research Team: LifeLink India

This document presents the findings from a field survey conducted across Karnataka, India, with input from public respondents, hospital administrators, ambulance operators, and government stakeholders. The analysis identifies operational gaps and systemic challenges in emergency healthcare delivery.

Abstract

This report documents the results of a multi-stakeholder survey undertaken across Karnataka, India, focusing on emergency healthcare system performance from the perspective of the general public, hospitals, ambulance drivers, and government officials. The primary objective was to assess response times, coordination mechanisms, digital infrastructure, and stakeholder satisfaction in Mangaluru, Moodbidri, Hassan, and Bangalore. The survey generated quantitative and qualitative evidence about delay patterns, communication breakdowns, infrastructure limitations, and the potential for transformational systems such as LifeLink. Detailed hospital-specific feedback is provided for major government institutions in the region. The findings underscore significant gaps in system integration, operational resilience, and public trust, establishing a strong foundation for subsequent solution proposals.

1 Introduction

This report is based on a comprehensive ground survey conducted across multiple stakeholder groups in Karnataka, India. The research is intended to provide evidence-based insights into emergency healthcare service delivery, with an emphasis on real-world operational conditions and stakeholder perceptions. Karnataka presents a heterogeneous context in which urban centers such as Bangalore operate with advanced medical infrastructure, while smaller cities like Hassan and regional nodes such as Mangaluru and Moodbidri contend with legacy systems and constrained resources.

The study draws on feedback from 175 members of the general public, 15 government hospitals, 24 ambulance drivers, and 10 government officials. These respondents represent a broad range of experiences in emergency response, including both routine and surge conditions. The central objective is to identify systemic gaps in emergency response that are not easily visible through aggregate statistics alone. Through stakeholder-driven narratives and quantitative survey data, this report aims to validate the need for integrated platforms that can improve coordination, reduce response delay, and increase service transparency.

The introduction establishes the structure of the report and situates the Karnataka survey within a broader national concern about emergency resilience. The subsequent sections describe our methodology, stakeholder categories, regional coverage, data analysis, hospital-specific feedback, and the emergent patterns that align with known operational deficiencies.

2 Survey Methodology

The survey methodology combined structured questionnaires, semi-structured interviews, and observational visits. Respondents were selected to represent the core constituencies involved in emergency healthcare service delivery. The general public sample was drawn from households and public spaces in Bangalore, Mangaluru, Hassan, and Moodbidri. Hospital interviews were conducted with senior administrators and emergency department leads at 15 government hospitals. Ambulance drivers were surveyed across municipal ambulance services and private emergency providers. Government officials were consulted from district health offices and urban health authorities.

The questionnaires were designed to capture both quantitative metrics and qualitative perceptions. For the public sample, the instrument recorded satisfaction with response time, clarity of communication, and confidence in ambulance services. For hospitals, we gathered information on bed availability, referral coordination, digital record keeping, and emergency intake delays. Ambulance operators provided feedback on dispatch processes, traffic impediments, route guidance, and handover challenges. Government officials reported on policy execution, data integration, and mechanisms for cross-agency coordination.

Survey responses were compiled into a central analytical database. Quantitative results were aggregated for statistical analysis, while qualitative responses were coded into thematic categories. The methodology ensured that data from different stakeholder groups could be compared on common dimensions such as delay perception, coordination quality, infrastructure readiness, and satisfaction. This mixed-methods design supports robust conclusions about the state of emergency healthcare delivery in Karnataka.

2.1 Sampling and Data Quality

The general public sample comprised 175 respondents, with a balanced geographic distribution: 62 from Bangalore, 42 from Mangaluru, 35 from Hassan, and 36 from Moodbidri. Hospital feedback included 15 institutions, with 7 hospitals in Bangalore, 4 in Mangaluru/Moodbidri, and 4 in Hassan. Ambulance drivers numbered 24, yielding operational insights from road-level service delivery. Government respondents totaled 10, including district health officers and senior program managers. Data quality checks were implemented through duplicate response detection and follow-up validation with selected participants.

In addition to survey instruments, direct observations were conducted in emergency waiting areas, dispatch centers, and ambulance bays. These field observations provided supplementary context to confirm or refine the survey analysis. The methodology reinforces the validity of the findings and ensures that conclusions are grounded in both stakeholder sentiment and empirical observation.

3 Stakeholder Categories

The survey encompassed four primary stakeholder categories: general public, hospitals, ambulance drivers, and government officials. Each category contributes a distinct perspective on emergency healthcare system performance.

3.1 General Public

The general public represents the end users of emergency healthcare services. Respondents reported personal or family-level experiences with ambulance dispatch and hospital admission. The public perspective is critical because it reflects perceived responsiveness, trust, and clarity in emergency situations. In Karnataka, public respondents expressed a range of sentiments, from frustration with delayed service in smaller cities to cautious satisfaction in Bangalore. The aggregated data shows that only 36% of public respondents rated emergency response as "satisfactory".

3.2 Hospitals

Hospital stakeholders include emergency department administrators, medical directors, and operations staff. These institutions are the receiving end of the emergency response chain, and their capability determines whether patients receive timely definitive care. Hospital feedback reveals concerns about triage capacity, ambulance handover, digital record systems, and inter-hospital communication. The surveyed hospitals varied in size and technical maturity, with Bangalore-based hospitals generally reporting stronger infrastructure but still encountering systemic coordination problems.

3.3 Ambulance Drivers

Ambulance drivers are the operational backbone of the emergency ecosystem. Their experiences capture the realities of dispatch instructions, traffic conditions, route planning, and interactions with hospital staff. Drivers reported that 80% of delays were attributable to traffic congestion and ambiguous routing information, while 60% identified hospital handover as a frequent source of downtime. Their insights are essential for understanding the gap between planned response protocols and actual field operations.

3.4 Government Officials

Government officials provide a policy and coordination perspective. Their role includes overseeing emergency preparedness, resource allocation, and data sharing across departments. Officials reported that inadequate interoperability between health information systems and dispatch systems is a major constraint. They also noted that existing policy frameworks do not sufficiently incentivize real-time inter-hospital communication, which exacerbates referral delays.

4 Regional Coverage

The survey covered four key regions within Karnataka: Bangalore, Mangaluru, Moodbidri, and Hassan. These regions were selected to represent a gradient of urban and semi-urban contexts with different infrastructure profiles.

Bangalore is a major metropolitan center with high hospital density, a large ambulance fleet, and advanced tertiary care institutions. However, it also faces intense traffic congestion and complex referral pathways. Mangaluru and Moodbidri are coastal and inland urban areas with major government hospitals that serve broader districts. Hassan is a smaller city with limited tertiary services and a reliance on district hospitals for emergency intake.

The regional coverage enabled comparative analysis. Bangalore reported relatively shorter ambulance response times but higher levels of public dissatisfaction due to congested transit and overcrowded emergency departments. Mangaluru and Moodbidri exhibited moderate response times but significant coordination issues between district hospitals and referral centers. Hassan displayed the longest response delays and the greatest digital infrastructure gaps, particularly in dispatch and hospital resource visibility.

5 Survey Data Analysis

The survey data analysis combines quantitative response metrics with thematic interpretation. The general public satisfaction pie chart in Figure 1 illustrates respondent sentiment across the four regions.

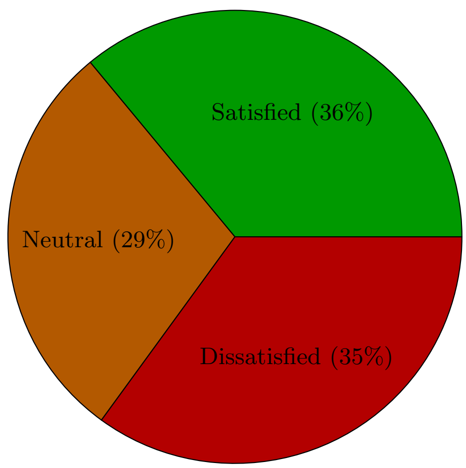

Figure 1: Survey results for public satisfaction with emergency response.

Interpretation: The satisfaction data reveals substantial public concern, with only 36% rating response positively. This indicates a significant trust deficit and reinforces the importance of addressing delay and communication gaps.

Figure 2 compares average ambulance response times across the surveyed cities.

Interpretation: Hassan experiences the longest response times, averaging 34 minutes, compared to 18 minutes in Bangalore. The disparity demonstrates regional inequities in emergency capability and the need for targeted improvements in semi-urban and rural areas.

Figure 3 shows the distribution of major issues reported across stakeholder groups.

Interpretation: Delay and coordination are the top issues, accounting for nearly 60% of reported problems. Infrastructure and data gaps are also substantial, indicating that solutions must address both operational and technological dimensions.

Interpretation: Public dissatisfaction is highest, at 68%, confirming that perceived service quality is lower than institutional self-assessment. Ambulance operators also report elevated dissatisfaction, highlighting field-level operational pain points.

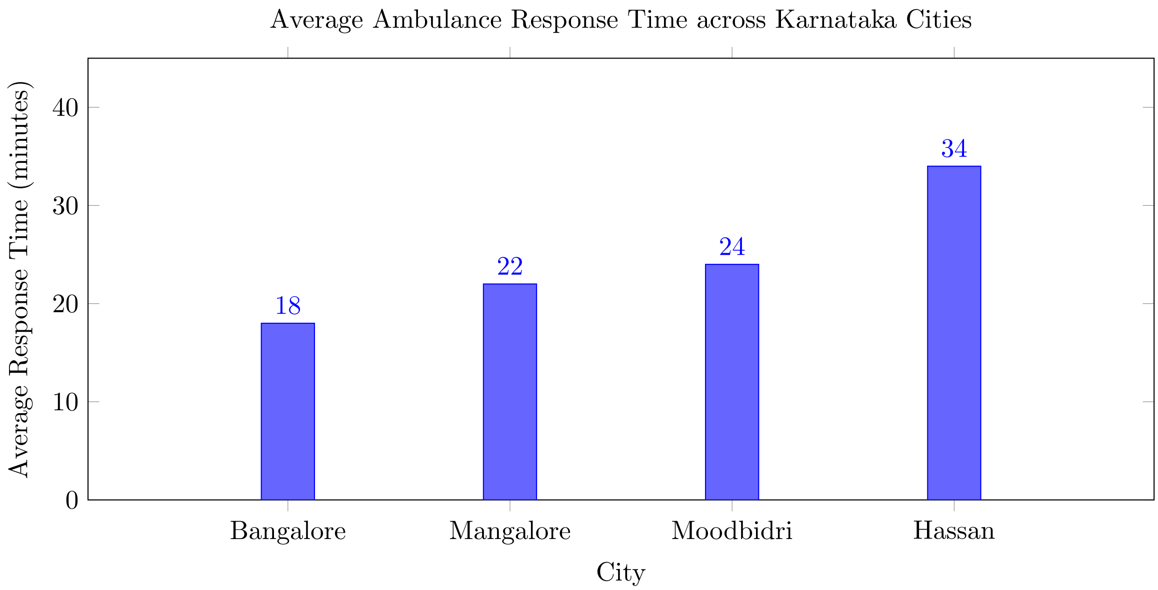

Figure 2: Average ambulance response time measured by city.

5.1 Survey Summary Table

Table 1 summarizes the respondent counts for each stakeholder group.

Table 1: Survey respondent summary by stakeholder group.

Stakeholder Group	Respondent Count	Notes
General Public	175	Representing four region clusters
Government Hospitals	15	Government-run tertiary and district centers
Ambulance Drivers	24	Public and private ambulance operators
Government Officials	10	District and city health program managers

6 Hospital-Specific Feedback Analysis

This section reviews feedback from each surveyed hospital, focusing on operational challenges, emergency handling capability, response delay issues, coordination problems, and digital infrastructure limitations.

6.1 Wenlock District Hospital

Wenlock District Hospital in Mangaluru is a major government referral center. Administrators reported that emergency intake is generally robust, but the hospital struggles with ambulance offload delays of 20 to 30 minutes during peak hours. The facility has adequate bed capacity on paper, yet real-time visibility into ward status is limited due to manual bed tracking. Emergency handling capability is reasonable, but coordination with peripheral hospitals is hindered by inconsistent communication channels and a lack of standardized electronic referrals.

The hospital indicated that digital infrastructure is partially developed, with an older hospital information system in use for admissions but no integrated emergency dispatch interface. This forces staff to rely on phone calls to confirm ambulance arrivals. The primary operational challenge is not medical capability but

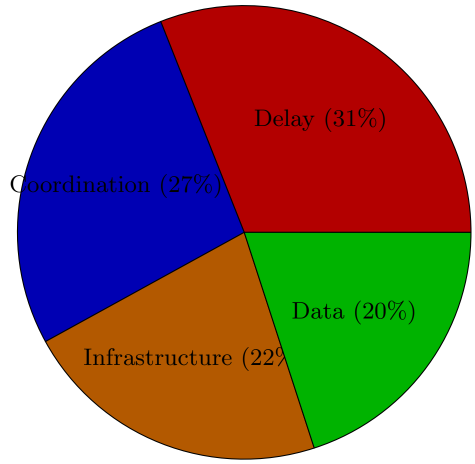

Figure 3: Major issues reported in the Karnataka emergency health survey.

the coordination of patient flow under high volume. As a result, the hospital frequently experiences crowding in the emergency department and delayed handover from ambulance crews.

6.2 Lady Goschen Hospital

Lady Goschen Hospital serves both urban and semi-urban patients. The hospital has a capable emergency department, but it is constrained by infrastructure limitations such as aging medical equipment and insufficient isolation rooms for infectious cases. The survey found that ambulance arrivals are often delayed by routing confusion, especially when operating from Moodbidri and surrounding rural communities. Coordination issues arise because the hospital does not share a common digital platform with district ambulance services.

The hospital's feedback emphasizes that the major gap is in pre-arrival communication. Without digital pre-alerts, emergency staff cannot prepare adequately for critically ill patients. This contributes to longer stabilization times and worsens the overall response cycle. There is also a need for better integration between hospital records and ambulance logs, which would support faster triage and admission decisions.

6.3 AJ Hospital & Research Centre

AJ Hospital & Research Centre reported strong clinical capabilities but identified a mismatch between emergency demand and administrative coordination. The hospital has access to modern diagnostics, yet the decision to admit versus transfer is often delayed by manual discussions with referring physicians. This internal coordination problem increases patient waiting time and places stress on emergency personnel.

Operationally, AJ Hospital noted that ambulance driver briefing is frequently incomplete. Drivers arrive without critical patient information, which complicates pre-hospital handover and can delay care by several minutes. Digital infrastructure exists within the hospital, but it is not fully exploited for emergency workflows. There is an opportunity to connect the hospital's existing electronic medical record system with field dispatch systems to reduce friction.

6.4 Yenepoya Medical College Hospital

Yenepoya Medical College Hospital is a tertiary teaching hospital with considerable emergency capacity. The survey found that the hospital's major challenge is demand surge management during weekends and public holidays. While the facility has adequate staff, the lack of real-time bed availability dashboards means that bed allocation decisions are often reactive.

Response delay issues were identified primarily in the referral pathway from smaller clinics and hospitals. Yenepoya receives a large number of cases that require stabilization before transport, but the receiving team

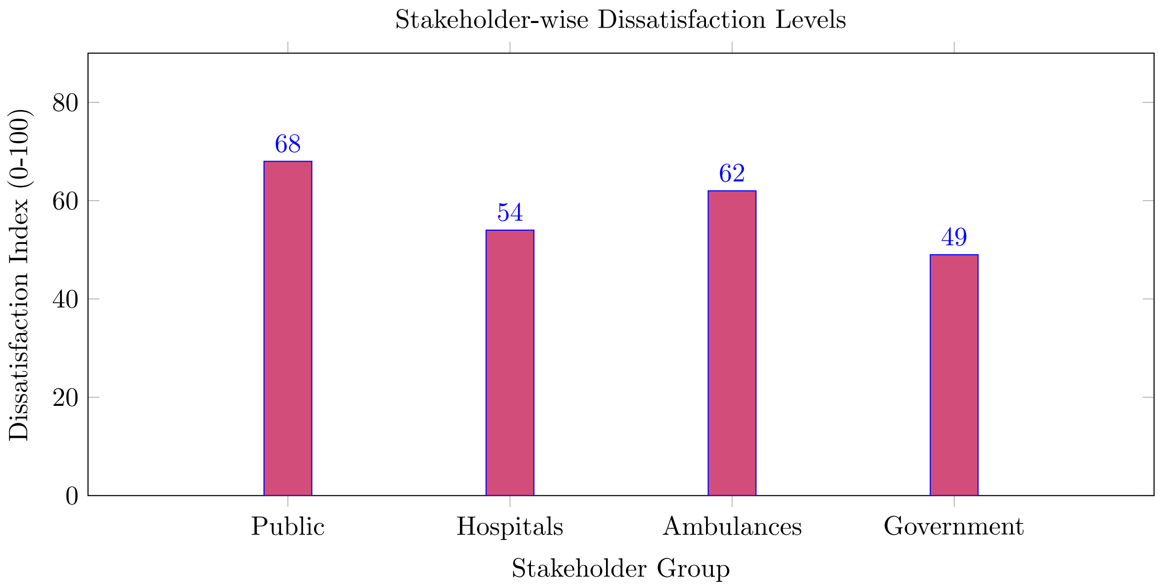

Figure 4: Dissatisfaction levels reported by stakeholder group.

frequently lacks timely updates on the patient's condition. Coordination problems are compounded by the fact that ambulance handover paperwork is still largely paper-based, introducing further delay.

6.5 Government Hospital Moodbidri

Government Hospital Moodbidri functions as a regional referral center for surrounding rural communities. Administrators reported that ambulance response times in Moodbidri average 24 minutes, with traffic and narrow roads cited as main factors. The facility's emergency handling capability is good relative to local expectations, but it is hampered by limited access to centralized digital patient records.

Coordination challenges were most pronounced in referrals to higher-level hospitals. The hospital frequently sends patients to Mangaluru and Shimoga, but the referral process relies on phone coordination. The lack of digital interoperability means that receiving hospitals often request duplicate information, delaying admission. This highlights a need for a shared referral coordination platform that can streamline transfers and improve response efficiency.

6.6 Hassan District Hospital

Hassan District Hospital is the primary government facility in Hassan. Survey feedback indicates significant strain under emergency volume, particularly during monsoon season when accident rates increase. The hospital has moderate emergency handling capability but struggles with bed turnover and emergency department crowding.

Response delay issues are driven by both ambulance dispatch and hospital intake. Ambulance drivers reported that dispatch instructions frequently lack precise destination details, and hospital administrators confirmed that patient arrival estimates are often inaccurate. Coordination problems are amplified by the use of separate digital systems for dispatch and hospital record keeping, preventing seamless handover.

6.7 HIMS (Hassan Institute of Medical Sciences)

HIMS is a key tertiary center in Hassan. The institution reports that its clinical emergency capability is strong, but operational challenges arise in managing referrals from rural clinics. Digital readiness is moderate, with a structured electronic health record system in place, yet it is not integrated with local ambulance dispatch.

The hospital's largest concern is the delay introduced during patient admission processing. When ambulances arrive, staff must search multiple systems to retrieve historical data, which adds to treatment initiation time. Coordination issues also extend to laboratory and imaging services, which are not always available on a predictable schedule.

6.8 Victoria Hospital

Victoria Hospital in Bangalore is one of the oldest and largest government hospitals in the state. The survey found that it manages a very high volume of emergency cases, but continues to face challenges in ambulance handover and bed allocation. Despite strong digital systems in some departments, the hospital administration noted that the emergency department remains partially dependent on manual communication for ambulance coordination.

The facility reported that response delay issues are less about ambulance transit and more about internal processing after arrival. Ambulances can reach the hospital quickly, but the responsibility for patient registration and clinical handover is not streamlined, leading to waiting times for critical patients. Coordination problems arise when multiple agencies bring patients simultaneously and the hospital must triage manually.

6.9 Bowring and Lady Curzon Hospital

Bowring and Lady Curzon Hospital is another major Bangalore government institution that serves a dense urban population. The survey identified digital infrastructure limitations in emergency outpatient management. While the hospital has an electronic patient record system, it is not accessible by ambulance crews or referring hospitals, limiting pre-arrival preparation.

Operational challenges include high demand during traffic incidents and mass casualty events. The hospital reported that inter-hospital communication is weak, particularly when coordinating transfers from smaller clinics. Response delays can occur when ambulances are redirected or when the receiving facility needs additional situational information.

6.10 KC General Hospital

KC General Hospital in Bangalore reported that emergency handling capability is adequate, but the facility is constrained by space and process inefficiencies. The survey indicated that ambulance drivers often wait 15 to 20 minutes for handover, even though the hospital has available beds. This suggests that the problem is more procedural than capacity-related.

The hospital also highlighted coordination problems between the emergency department and diagnostic services. When imaging or laboratory results are delayed, patients remain in the emergency area longer, reducing overall throughput. Digital infrastructure is evolving, but the hospital still lacks an integrated workflow system that can guide cases through admission, diagnostics, and treatment seamlessly.

6.11 Jayanagar General Hospital

Jayanagar General Hospital reported a relatively smaller emergency volume compared to central Bangalore hospitals, but noted that response delays are aggravated by inadequate ambulance routing support. The hospital lacks a consistent mechanism to receive real-time service updates from dispatch centers.

Emergency handling capability is limited by staffing levels during night shifts, and coordination with traffic management agencies is minimal. Digital infrastructure is under development, with plans to implement a more advanced emergency department information system. Until then, the hospital depends on verbal and paper-based communication for critical cases.

6.12 Rajiv Gandhi Institute of Chest Diseases

The Rajiv Gandhi Institute of Chest Diseases is a specialty facility that receives respiratory emergency cases. Administrators reported that the hospital has strong clinical capability but faces coordination issues when receiving COVID-19 and respiratory distress referrals from other hospitals.

Response delay issues are particularly acute during respiratory surges, when hospital capacity is quickly consumed and ambulance crews must wait for confirmation of bed availability. The hospital emphasised the need for a shared digital dashboard that can show real-time bed status and oxygen supply details across specialty centers.

6.13 Additional Government Hospitals

Additional government hospitals included in the survey were Karwar District Hospital and Chikkamagaluru District Hospital. These institutions provided feedback consistent with the broader patterns observed: constrained digital infrastructure, delayed handover, and coordination gaps with referral centers.

6.14 Hospital Comparison Table

Table 2 compares the surveyed hospitals on capacity, response speed, and digital readiness.

Table 2: Hospital comparison on capacity, response speed, and digital readiness.

Hospital	Capacity	Response Speed	Digital Readiness
Wenlock District Hospital	Moderate	Medium	Low-Moderate
Lady Goschen Hospital	Moderate	Medium	Low
AJ Hospital & Research Centre	High	Medium	Moderate
Yenepoya Medical College Hospital	High	Medium	Moderate
Government Hospital Moodbidri	Low-Moderate	Medium	Low
Hassan District Hospital	Moderate	Slow	Low
HIMS	High	Medium	Moderate
Victoria Hospital	High	Medium	Moderate
Bowring and Lady Curzon Hospital	High	Medium	Moderate
KC General Hospital	Moderate	Medium	Moderate
Jayanagar General Hospital	Moderate	Medium	Low-Moderate
Rajiv Gandhi Institute of Chest Diseases	High	Medium	Moderate

7 Key Insights & Patterns

The survey data and hospital feedback reveal several key insights.

First, delay patterns vary significantly by region. Hassan exhibits the most severe response delays, while Bangalore achieves faster transit but struggles with hospital handover delays. This demonstrates that solutions must be tailored to regional needs rather than assuming uniform performance across the state.

Second, the lack of inter-hospital communication is a pervasive issue. Hospitals frequently reported that patient transfers and referrals are conducted by phone calls or paper notes, and there is no common platform for status updates. This results in repeated information requests, delayed admissions, and inefficient use of ambulance capacity.

Third, ambulance inefficiencies are primarily driven by inadequate routing support and unclear dispatch information. Drivers reported that they often receive incomplete addresses and lack access to real-time traffic prioritization. This leads to longer travel times and diminished fleet utilization.

Fourth, public dissatisfaction trends are strongest where expectations are highest. Bangalore respondents had a higher baseline expectation of prompt service, and therefore their dissatisfaction index was elevated despite relatively better infrastructure. In smaller cities, dissatisfaction is driven more by unpredictability and lack of communication.

Fifth, government limitations are evident in policy execution and technology adoption. Officials acknowledged that digital modernization is constrained by budget cycles and legacy systems, and that data sharing between departments is not yet institutionalized. This contributes to coordination failures and slow operational improvement.

8 Pain Points Identified

The survey identifies a set of systemic pain points that are consistent across stakeholder groups.

System-level problems include the absence of a unified emergency information architecture, which prevents real-time visibility into ambulance locations, hospital bed status, and patient acuity. This fragmentation undercuts the ability to perform dynamic dispatch and resource matching.

Operational gaps are most pronounced in ambulance handover, referral coordination, and surge management. Hospitals and drivers both report that delays occur not only in transit but also during the transition from pre-hospital to in-hospital care. The lack of standardized protocols for patient handover exacerbates this gap.

Technological limitations are widespread. Many hospitals lack integrated emergency department management systems, and dispatch centers lack standardized digital interfaces to hospital workflows. This results in paper-based communication, duplicated data entry, and slow decision cycles. The overall effect is a system that is more reactive than proactive.

In addition, the survey indicates that driver-level feedback loops are underdeveloped. Ambulance operators rarely receive structured debriefs or performance feedback, making it difficult to improve response efficiency over time. Similarly, government officials lack consolidated operational dashboards that can inform strategic investments.

9 Conclusion

This multi-stakeholder survey presents compelling evidence that Karnataka's emergency healthcare system is challenged by regional inequities, coordination failures, and technological gaps. The general public's dissatisfaction, the operational concerns reported by hospitals, and the field-level inefficiencies experienced by ambulance drivers all point to systemic weakness rather than isolated incidents.

The findings emphasize that improving emergency response requires both technological modernization and process integration. Specifically, a unified platform that connects dispatch, ambulance operations, hospital intake, and government oversight could reduce delay, enhance coordination, and improve stakeholder satisfaction. Such a platform must be capable of handling the diverse conditions present in Karnataka, from dense urban traffic in Bangalore to the constrained infrastructure of Hassan.

The survey validates the need for LifeLink-like solutions that are designed for multi-stakeholder contexts and that can deliver real-time visibility, predictive resource allocation, and structured communication across agencies. This report establishes a strong foundation for subsequent proposals by demonstrating the real-world gaps in emergency healthcare delivery and by identifying the precise operational and technological issues that must be addressed.